

7	Scalar and vector quantities	Notes
	The addition and subtraction of coplanar vectors and the multiplication of a vector by a scalar. Components and resolved parts of a vector. Magnitude of a vector. Position vector. Unit vector. Use of vectors to establish simple properties of geometrical figures.	Knowledge of the fact that if $\alpha_1 \mathbf{a} + \beta_1 \mathbf{b} = \alpha_2 \mathbf{a} + \beta_2 \mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors, then $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$, is expected. Use of the vectors $\mathbf{i}$ and $\mathbf{j}$ will be expected. $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$ The ‘simple properties’ will, in general, involve collinearity, parallel lines and concurrency. Position vector of a point dividing the line AB in the ratio $m:n$ is expected.